JAMHURIYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

LINUX ASSIGNMENT

NAME: SAAFI ABDULLAHI MOHAMED

ID: C6240282

SUBMISSION DATE: 12 NOVEMBER 2026

CHAPTER 1

ENTRADOCTION LINUX AND UBUNTU

1.whoami

- **Whoami:** - It displays the username of the person currently logged into the terminal.

Syntax: Whoami [**OPTION**]

OPTIONS:

- --**help**: Displays help information and an explanation of the command.
- --**version:** Displays the version of the command.

```
saffah@saffah:~$ whoami
saffah
saffah@saffah:~$ whoami --help
Usage: whoami [OPTION]...
Print the user name associated with the current effective user ID.
Same as id -un.

      --help     display this help and exit
      --version  output version information and exit

GNU coreutils online help: <http://www.gnu.org/software/coreutils/>
Full documentation at: <http://www.gnu.org/software/coreutils/whoami>
or available locally via: info '(coreutils) whoami invocation'
saffah@saffah:~$ whoami --version
whoami (GNU coreutils) 8.28
Copyright (C) 2017 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Written by Richard Mlynarik.
saffah@saffah:~$
```

2.pwd

- **Pwd:** stands for **"print working directory".**
  It shows the folder you are currently in, using the full path that starts from the root (/).

  Syntax: pwd [option]

  Options:

  - -L: shows the "logical" path (follows symbolic links) This is the default.
  - -P: shows the actual ("physical") path, without following symbolic links.
  - --help: shows help.

```
saffah@saffah:~$
saffah@saffah:~$ pwd
/home/saffah
saffah@saffah:~$ pwd -l
bash: pwd: -l: invalid option
pwd: usage: pwd [-LP]
saffah@saffah:~$ pwd -p
bash: pwd: -p: invalid option
pwd: usage: pwd [-LP]
saffah@saffah:~$ pwd --help
pwd: pwd [-LP]
    Print the name of the current working directory.

    Options:
      -L        print the value of $PWD if it names the current working
                directory
      -P        print the physical directory, without any symbolic links

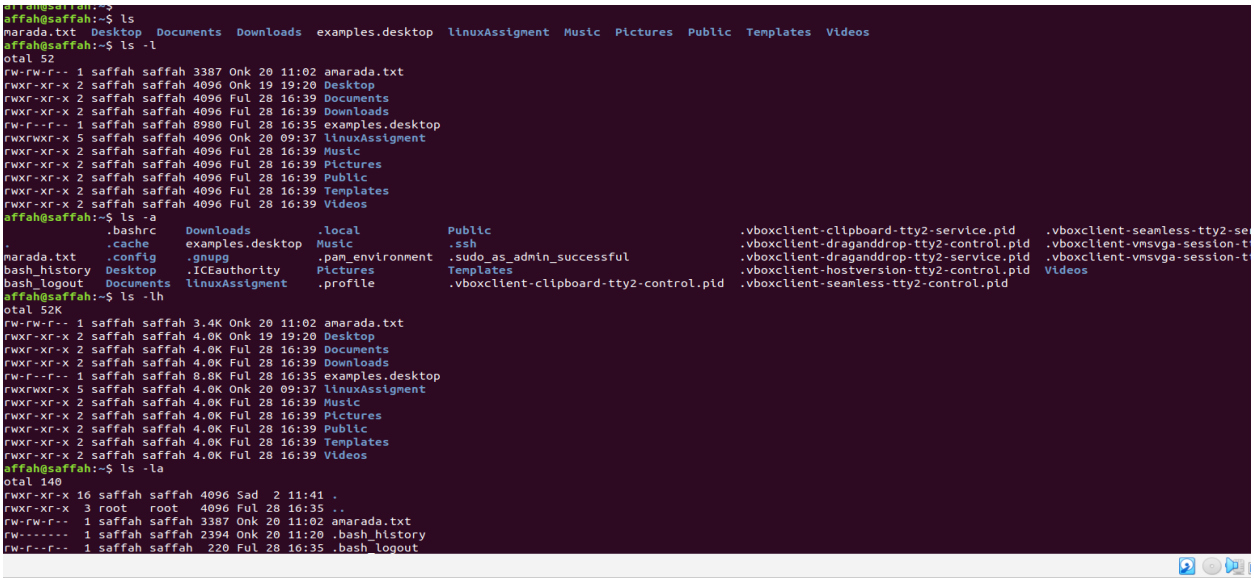
    By default, `pwd' behaves as if `-L' were specified.

    Exit Status:
    Returns 0 unless an invalid option is given or the current directory
    cannot be read.
saffah@saffah:~$
```

3.ls

- **ls:** stands for **"list".** It lists the files and folders inside the folder you are currently in.

  Syntax: ls [OPTION] [FILE/DIRECTORY]
  Options:
  - -l: long listing (permissions, owner, size, date).
  - -a: shows hidden files (those that start with a dot .).
  - -h: makes sizes easy to read (KB, MB); used together with -l as -lh.
  - -t: sorts by time modified.
  - -r: reverses the sort order.
  - -R: shows the contents of subfolders too (recursive).
  - --help: shows help.



4.cat/etc/os-release

- **Cat:** reads a file and displays its contents in the terminal. /etc/os-release is a file that contains information about the distribution (the Ubuntu name and version).

  Syntax: cat [OPTION] [FILE]

  Options:

  - -n: adds a number to every line.
  - -b: numbers only the non-blank lines.
  - -A: shows hidden characters.
  - --help: shows help.



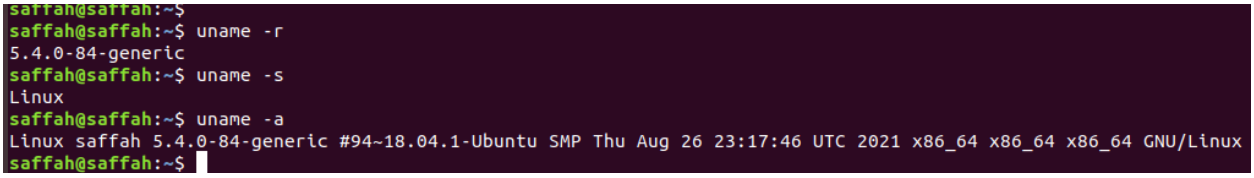
5.uname -r

- **Uname: (unix name)** displays system information.
  The -r (**release**) option shows the release of the Linux kernel that is currently running.

  Syntax: uname [OPTION]

  Options:

  - -r: shows the release of the kernel.
  - -s: shows the kernel name (Linux; on macOS it returns Darwin).
  - -n: shows the computer's name (hostname).
  - -m: shows the hardware type (for example x86_64).
  - -a: shows all the information at once.
  - --help: shows help.



Chapter 2

LINUX SYSTEM ARCHITECTURE

COMMAND LINE AND FILE MANAGMENT

1.echo

- **echo:** prints text or the value of a variable to the terminal (for example: echo $PATH, echo $USER).

  Syntax: echo [OPTION] [STRING]

  OPTIONS:
  - -n: does not add a new line at the end.
  - -e: interprets backslash escapes such as \n.
  - " " (double quotes): preserve spaces, but $ is still expanded (variables are still replaced with their values).
  - ' ' (single quotes): print everything exactly as written.



2.cd

- **Cd:** moves you to another directory (change directory). It is a built-in command of Bash.

  Syntax: cd [DIRECTORY]

  Options:

  - cd /var/log: an absolute path, the direct path starting from the root (/).
  - cd Documents: a relative path, goes to the Documents folder inside the directory you are currently in.
  - cd ..: goes up one level (parent directory).
  - cd or cd ~: takes you back to your home directory.



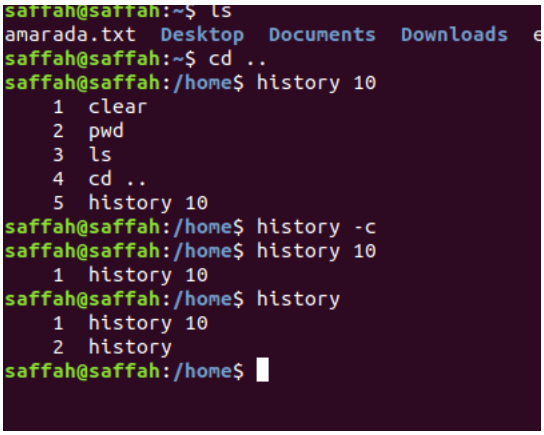
3.history

- **history:** shows the list of previous commands you typed in the terminal. It is stored in ~/.bash_history.

  Syntax: history [OPTION]

  OPTIONS:

  - up arrow / down arrow: the arrow keys that bring back previous commands.
  - Ctrl + R: search; type part of the command to find it.
  - -c: clears the history.
  - --help: shows help and a brief description of the command.

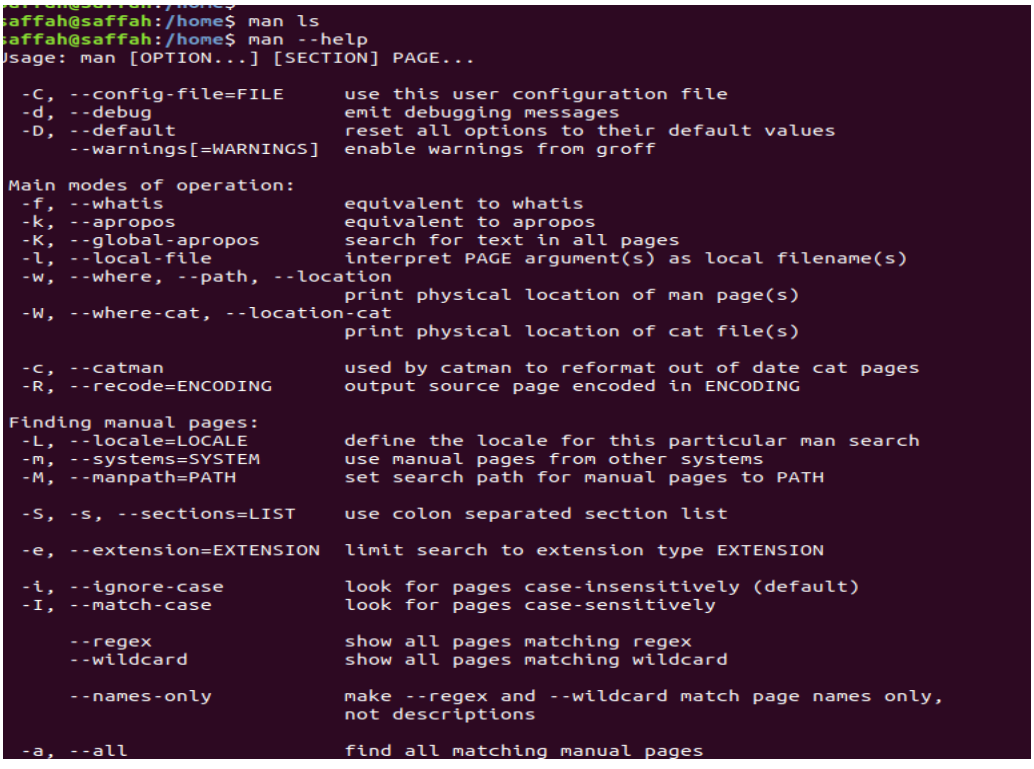


4.man

- **Man:man [SECTION] COMMAND**

  OPTIONS:

  - Space: next page.
  - b: previous page.
  - /word: search for a word.
  - n: next search result.
  - q: quit.
  - --help: shows help and a brief description of the command.

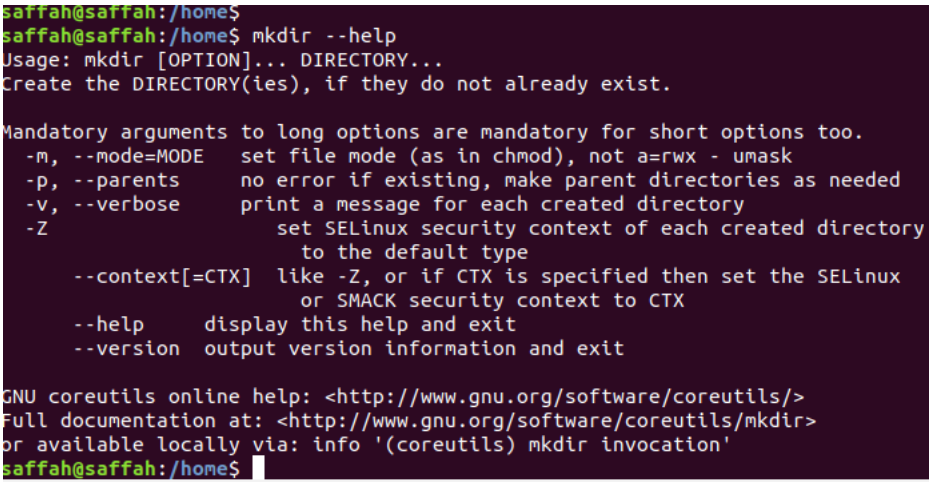


5.help

- --**help:** an option that many commands have. It prints a short, simple summary on the screen about how to use the command.

Syntax: COMMAND –help

Tusaale: mkdir –help

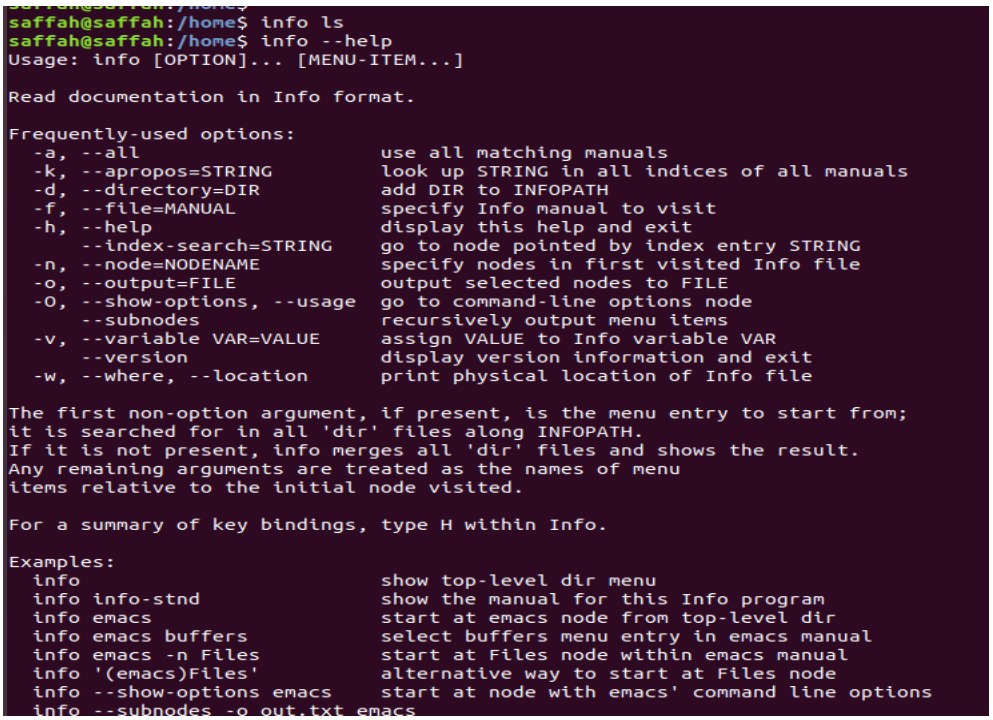


6.info

- **info:** opens the detailed documentation, divided into chapters, that comes with most GNU tools.

  Syntax: info [COMMAND]

  OPTIONS:
  - Space: scroll down.
  - q: quit.
  - --help: shows help and a brief description of the command.



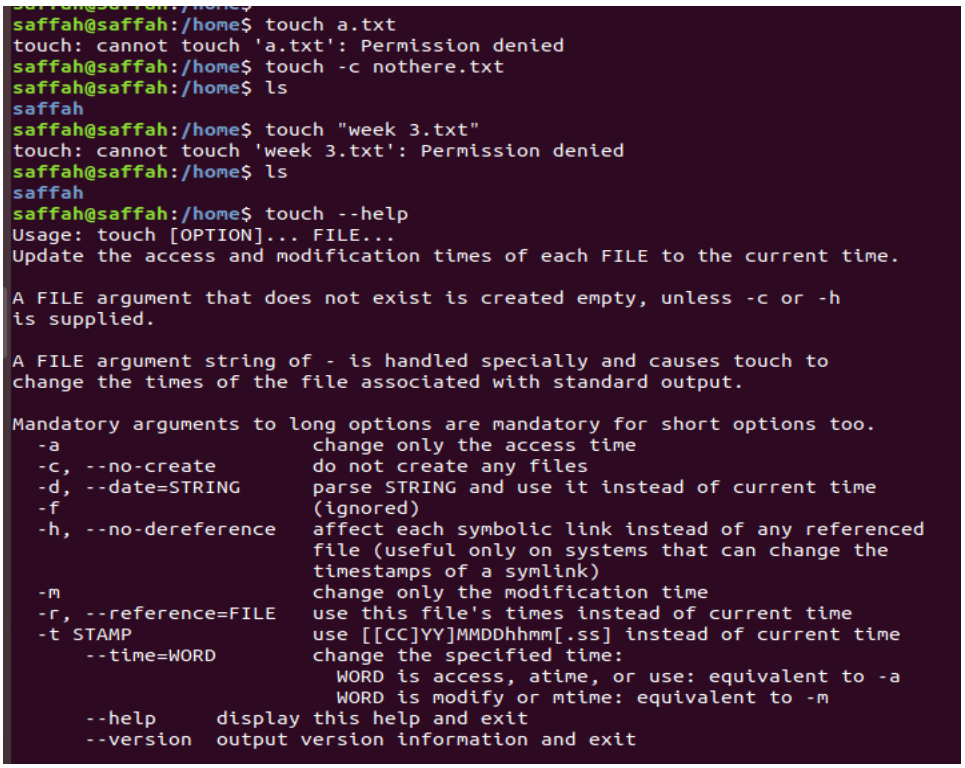
7.touch

- **touch:** creates an empty file, or updates the time of a file that already exists.

Syntax: touch [OPTION] FILE...

OPTIONS:

- -c: do not create the file if it does not already exist.
- "file name": if the name contains a space, put it in quotes so it is treated as one file.
- --**help**: shows help and a brief description of the command.
- --**version**: shows the version of the command.



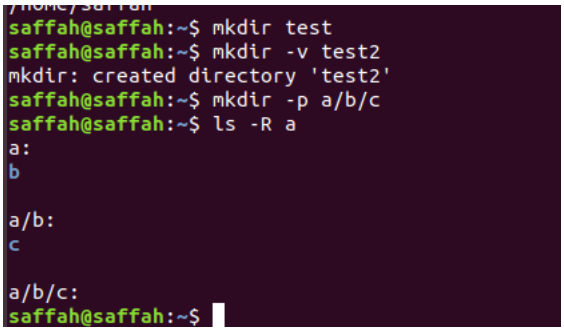
8.mkdir

mkdir: creates a new directory (make directory).

Syntax: mkdir [OPTION] DIRECTORY...

OPTIONS:

- -p: create missing parent directories.
- -v: print a message every time a directory is created.
- --**help**: shows help and a brief description of the command.
- --**version**: shows the version of the command.



9.cp

• **cp:** copies a file or directory. The original stays in place, so you end up with two copies.

Syntax: cp [**OPTION**] SOURCE DEST

OPTIONS:

- -r: copy a whole directory and everything inside it.
- -i: ask before overwriting.
- -v: show what is being copied.
- --help: shows help and a brief description of the command.
- --version: shows the version of the command.

```
saffah@saffah:~$ touch a.txt
saffah@saffah:~$ cp a.txt b.txt
saffah@saffah:~$ cp -v a.txt c.txt
'a.txt' -> 'c.txt'
saffah@saffah:~$ cp -i a.txt b.txt
cp: overwrite 'b.txt'? n
saffah@saffah:~$ mkdir d
saffah@saffah:~$ touch d/x.txt
saffah@saffah:~$ cp -r d d2
saffah@saffah:~$ ls
a  amarada.txt  a.txt  b.txt  c.txt  d  d2  Desktop  Documents  Downloads  examples.desktop
saffah@saffah:~$ ls d2
x.txt
saffah@saffah:~$ cp --help
Usage: cp [OPTION]... [-T] SOURCE DEST
  or:  cp [OPTION]... SOURCE... DIRECTORY
  or:  cp [OPTION]... -t DIRECTORY SOURCE...
Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.

Mandatory arguments to long options are mandatory for short options too.
  -a, --archive                same as -dR --preserve=all
      --attributes-only        don't copy the file data, just the attributes
      --backup[=CONTROL]       make a backup of each existing destination file
  -b                           like --backup but does not accept an argument
      --copy-contents          copy contents of special files when recursive
  -d                           same as --no-dereference --preserve=links
  -f, --force                  if an existing destination file cannot be
                                 opened, remove it and try again (this option
                                 is ignored when the -n option is also used)
  -i, --interactive            prompt before overwrite (overrides a previous -n
                                 option)
```

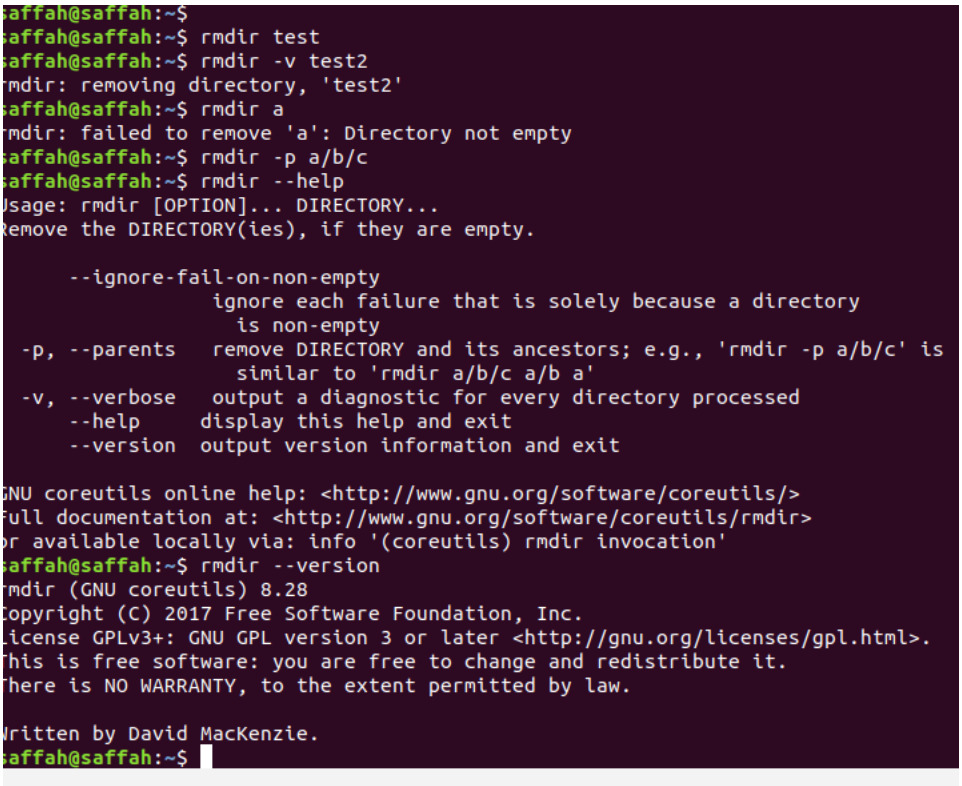
10.rmdir

- **rmdir:** removes a directory, but only if it is empty.

Syntax: rmdir [OPTION] DIRECTORY...

OPTIONS:

- -p: also remove empty parent directories.
- -v: show what is being removed.
- --**help**: shows help and a brief description of the command.
- --**version**: shows the version of the command.



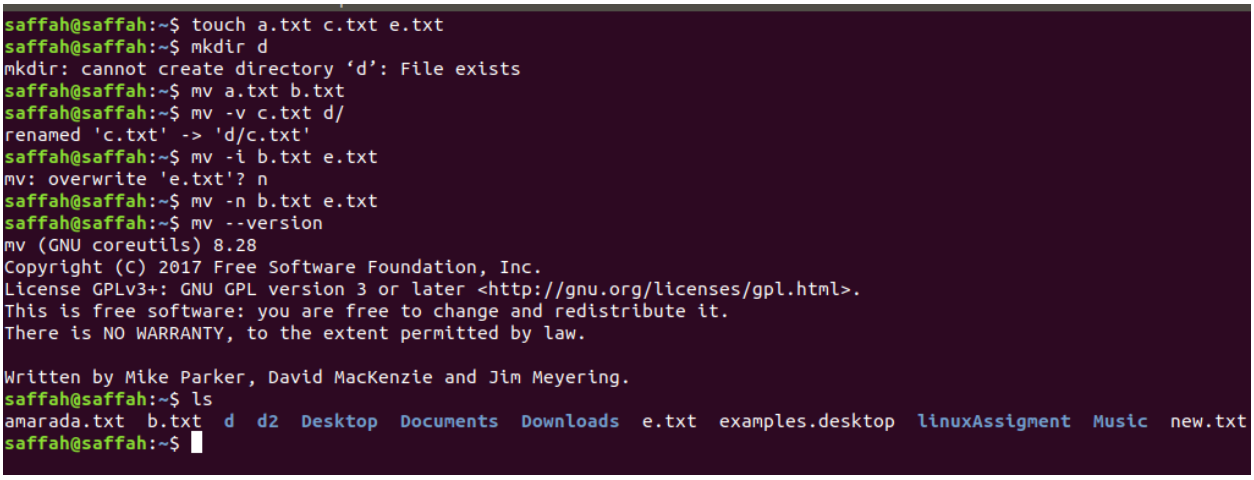
11.mv

- **`mv:`** Moves (**move**) a file or directory, or changes its name (renames it).

Syntax: mv [OPTION] SOURCE DEST

OPTIONS:

- -i: Asks for confirmation before overwriting
- -n: Does not overwrite an existing file
- -v: Shows what was moved (verbose)
- --help: Displays help and information about the command
- --version: Displays the version of the command



12.rm

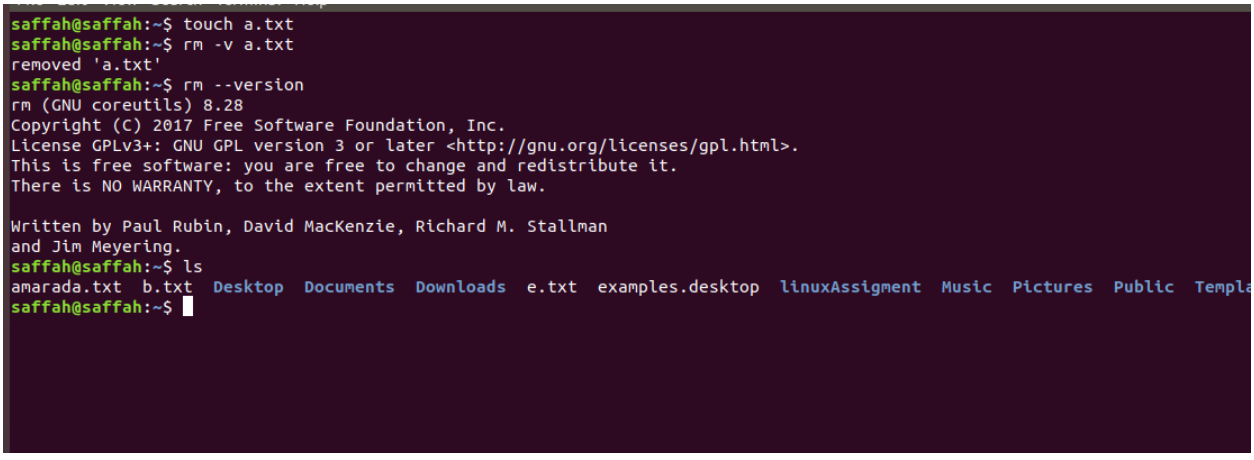
- **rm:** Used to permanently remove (remove) files or directories.

There is no Recycle Bin and no undo.

Syntax: rm [OPTION] FILE...

OPTIONS:

- -i: Asks before each removal; you answer y or n
- -r: Removes a directory and everything inside it
- -f: Force, with no prompts and no errors shown
- -rf: A dangerous combination, so be careful with it
- --help: Displays help and information about the command
- --version: Displays the version of the command



13.export

- **export:** Marks a variable as part of the environment, so programs launched from the shell can inherit it. A variable is defined as NAME=value (no spaces allowed).

Syntax: export [NAME[=value]]

OPTIONS:

- **NAME**=value: Creates a shell variable
- $**NAME**: The value of the variable
- -p: Lists all variables that have been exported

```
saffah@saffah:~$
saffah@saffah:~$ COURSE=linux
saffah@saffah:~$ echo $COURSE
linux
saffah@saffah:~$ bash
saffah@saffah:~$ echo $COURSE

saffah@saffah:~$ exit
exit
saffah@saffah:~$ export COURSE
saffah@saffah:~$ bash
saffah@saffah:~$ echo $COURSE
linux
saffah@saffah:~$ exit
exit
saffah@saffah:~$ export -p
declare -x CLUTTER_IM_MODULE="xim"
declare -x COLORTERM="truecolor"
declare -x COURSE="linux"
declare -x DBUS_SESSION_BUS_ADDRESS="unix:path=/run/user/1000/bus"
declare -x DESKTOP_SESSION="ubuntu"
declare -x DISPLAY=":0"
declare -x GDMSESSION="ubuntu"
declare -x GJS_DEBUG_OUTPUT="stderr"
declare -x GJS_DEBUG_TOPICS="JS ERROR;JS LOG"
declare -x GNOME_DESKTOP_SESSION_ID="this-is-deprecated"
declare -x GNOME_SHELL_SESSION_MODE="ubuntu"
declare -x GNOME_TERMINAL_SCREEN="/org/gnome/Terminal/screen/6337151a_f160_45ba_a03e_98de3254c3dd"
declare -x GNOME_TERMINAL_SERVICE=":1.64"
declare -x GPG_AGENT_INFO="/run/user/1000/gnupg/S.gpg-agent:0:1"
declare -x GTK_IM_MODULE="ibus"
declare -x GTK_MODULES="gail:atk-bridge"
declare -x HOME="/home/saffah"
declare -x IM_CONFIG_PHASE="2"
declare -x LANG="en_GB.UTF-8"
declare -x LANGUAGE="en_GB:en"
declare -x LC_ADDRESS="om_KE.UTF-8"
declare -x LC_IDENTIFICATION="om_KE.UTF-8"
```

CHAPTER 3

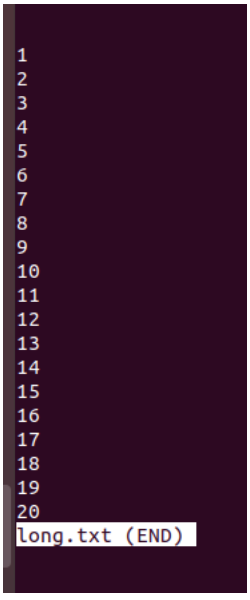
TEXT PROCESSING SEARCHING ARCHIVING AND EDITORS

1 less

- **less:** Used to view long files one page at a time.

  Syntax: less [OPTION] FILE
  OPTIONS:

  - -N: Shows line numbers
  - Space: Goes to the next page
  - b: Goes to the previous page
  - /word: Searches for a word
  - n: Goes to the next match
  - q: Quits the command



2.more

- **more**:Used to view long files one page at a time (simpler than less).

  Syntax: more [OPTION] FILE

  OPTIONS:

  - Space: Goes to the next page
  - Enter: Goes down one line
  - -d: Shows helpful messages
  - q: Quits



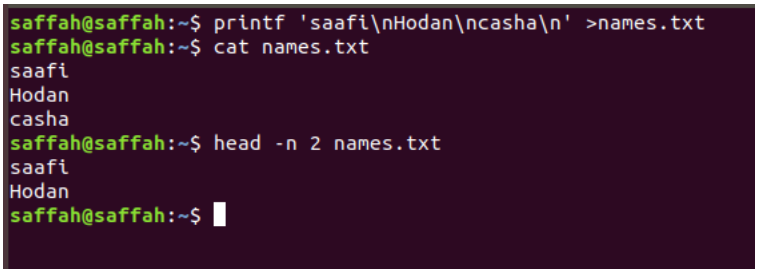
3. head

- **head**: Used to view the beginning of a file (default is 10 lines).

  Syntax: head [OPTION] FILE

  OPTIONS:

  - -n N: Shows the first N lines
  - -c N: Shows the first N bytes
  - -n -N: Shows everything except the last N lines



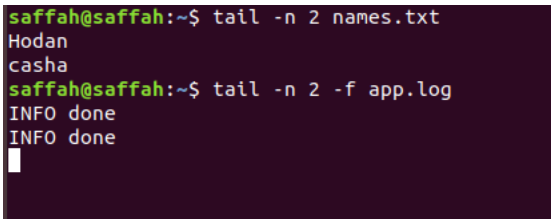
4.tail

- **tail**: It is used to view the end of a file (default 10 lines).

Syntax: tail [OPTION] FILE

OPTIONS:

- -n N: Shows the last N lines.
- -n +N: Shows the file starting from line N.
- -c N: Shows the last N bytes.
- -f: Follows the file in real time (Ctrl+C to stop it).



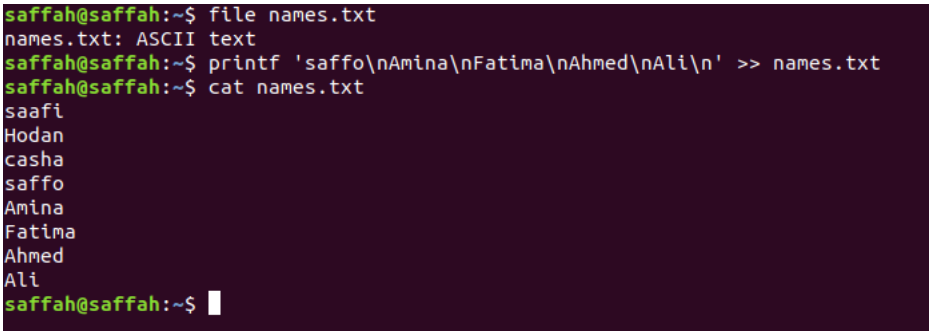
5.file

- **file**: It is used to identify the real type of a file.

Syntax: file [OPTION] FILE

OPTIONS:

- -b: Hides the file name (brief output).
- -i: Shows the MIME type.
- -L: Follows symbolic links.



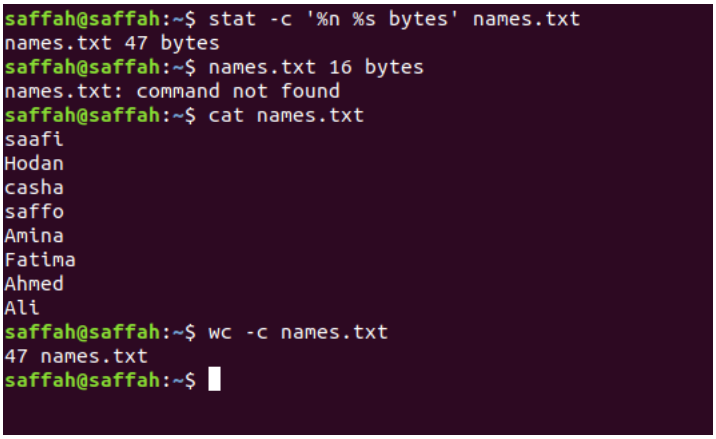
6.stat

- **stat**: It is used to view detailed information about a file (size, inode, permissions, time).

  Syntax: stat [OPTION] FILE

  OPTIONS:

  - -c FORMAT: Uses a custom format (%n name, %s size, %a permissions).
  - -f: Shows filesystem information.
  - -L: Follows links.



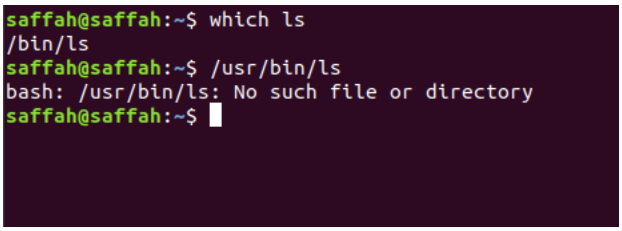
7.which

- **which**: It is used to find the location of an executable file in your PATH. Built-in commands like cd are not shown, because they are part of the shell and are not separate files.

Syntax: which [OPTION] COMMAND

OPTIONS:

- -a: Shows all the locations where the command is found, not just the first one.



8.type

- **type**: It shows how the shell interprets a command name.

A name can be one of four kinds:

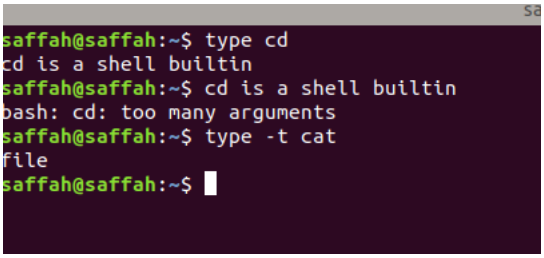
- builtin: part of the shell itself, like cd
- file: a separate program stored on disk, like cat
- alias: a shortcut you defined, like ll
- function: a small script defined inside the shell

This matters because the type decides how the command behaves. For example, which can't find built-ins like cd, but type can.

Syntax: type [-a] [-t] NAME

OPTIONS:

- -t: Prints only one word: builtin, file, alias, or function. This is useful in scripts.
- -a: Shows all the places where the name is found. A command can exist as an alias and as a file at the same time.



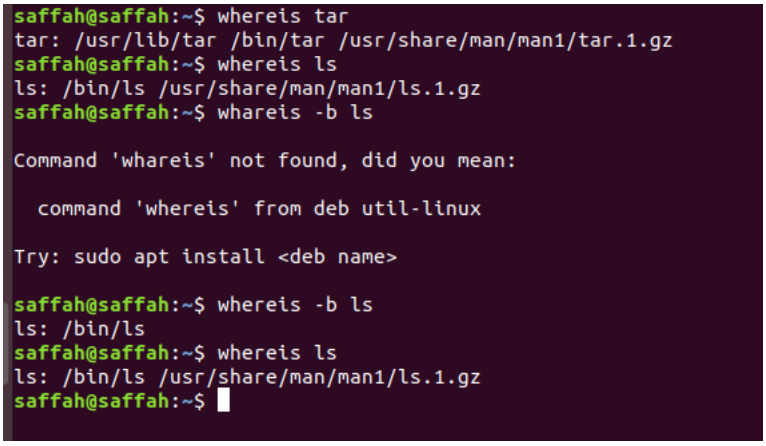
9.whereis

- **whereis**:It locates the binary, the source and the manual pages of a command.

  Syntax: whereis [OPTION] COMMAND

  OPTiONS:

  - -b: Shows only the binary.
  - -m: Shows only the manual.
  - -s: Shows only the source.



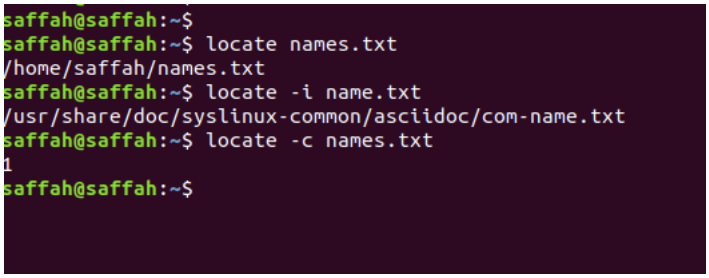
10.locate

Locate: It is used to quickly find files and directories.

Syntax: locate [OPTION] FILE

OPTIONS:

- -i: Ignores the difference between uppercase and lowercase.
- -n: Limits the number of results displayed.
- -c: Counts the results.

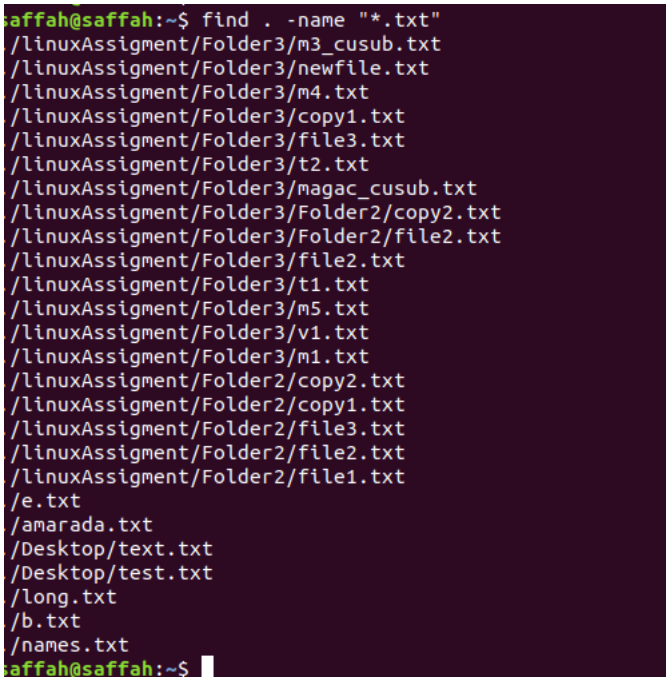


11.find

- **find**: It searches live for files and directories by name, type, size, and time.

Syntax: find [PATH] [EXPRESSION]
OPTIONS:

- -name 'pattern': Search by name (case-sensitive: uppercase and lowercase are treated differently).
- -iname: Same as -name, but it does not distinguish between uppercase and lowercase.
- -type f / d: f = file, d = directory.
- -size +N / -N: Larger / smaller (c = bytes, k, M, G).
- -mmin -N: Modified within the last N minutes.



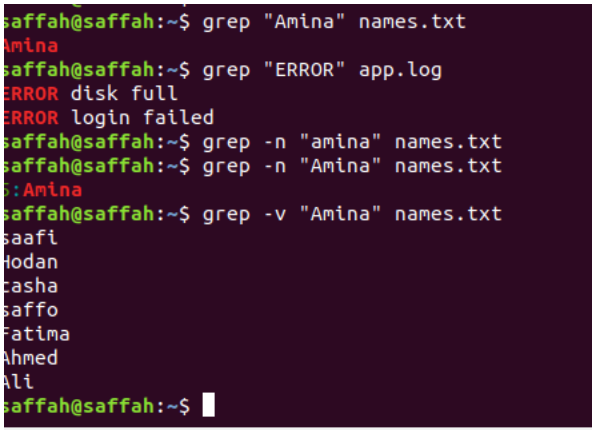
12.grep

Grep: It searches for a word or pattern inside files, and shows the lines that contain it.

Syntax: grep [OPTION] PATTERN [FILE]

OPTIONS:

- -n: Show the line number.
- -i: Ignore the difference between uppercase and lowercase.
- -v: Show the lines that do not match.
- -r: Search all directories and sub-directories.
- -c: Count the number of matching lines.
- -l: Show only the names of files that contain a match.



CHAPTER 4

PROCESS SERVICE AND SYSTEM MANAGMENT

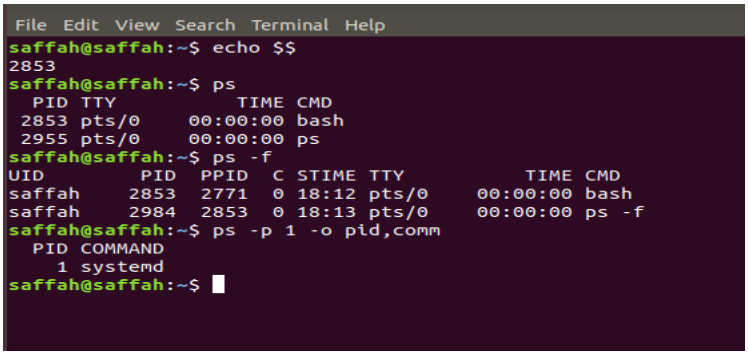
1.ps

Ps: It searches for a word or pattern inside files, and shows the lines that contain it.

Syntax: ps [OPTION] PATTERN [FILE]

OPTIONS:

- -n: Show the line number.
- -i: Ignore the difference between uppercase and lowercase.
- -v: Show the lines that do not match.
- -r: Search all directories and sub-directories.
- -c: Count the number of matching lines.
- -l: Show only the names of files that contain a match.



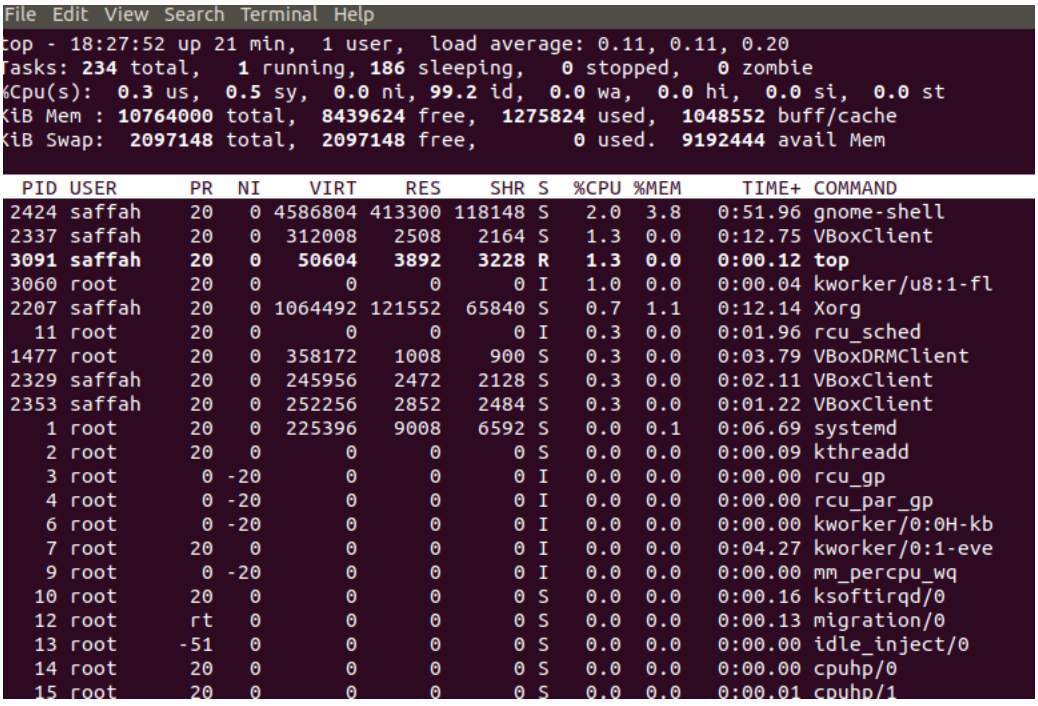
2.top

Top: It shows a live list of processes, refreshed every few seconds, busiest first.

Syntax: top [OPTION]

OPTIONS:

- -u **USER**: Show only that user's processes.
- -p **PID**: Watch one process.
- P (**key**): Sort by CPU.
- M (**key**): Sort by memory.
- k (**key**): Kill a process.
- r (keby): Renice a process.
- q (**key**): Quit.



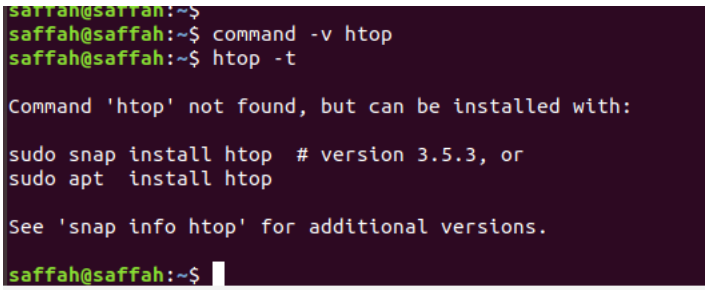
3.htop

Htop: A friendlier version of top with colour bars and menus (if installed). Check with: command -v htop

Syntax: htop [OPTION]

OPTIONS:

- -u **USER**: Only that user's processes.
- -t: Tree view.
- F6: Choose the sort column.
- F9: Send a signal to the selected process.
- F10 or q: Quit.



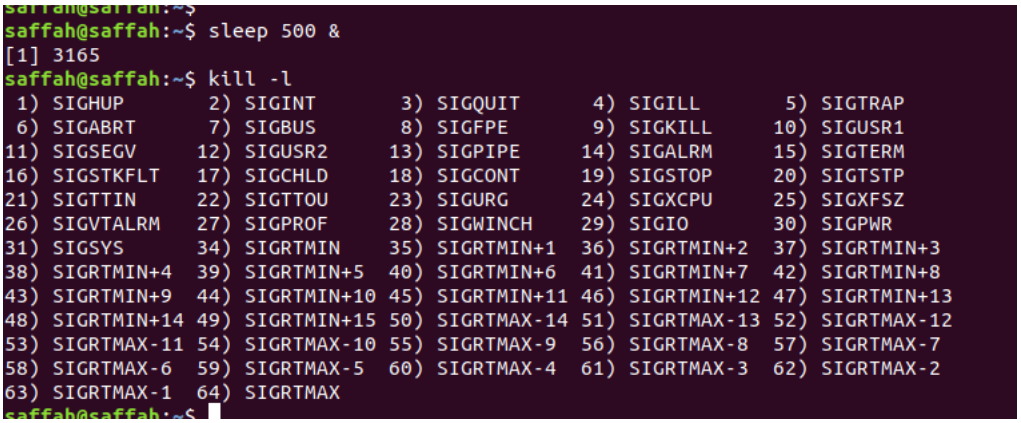
4.kill

- **Kill**:It sends a signal to one process by its PID. Without a signal name it sends SIGTERM (15), the polite request to finish.

  Syntax: kill [-SIGNAL] PID or kill %N

  OPTIONS:

  - (**none**): SIGTERM (15), please finish and exit.
  - -**TERM**: The same, written out.
  - -**HUP** (1): Hang up. Many daemons reload their configuration.
  - -**INT** (2): Interrupt, same as Ctrl+C.
  - -9 / -KILL: End now. It cannot save or clean up. Last resort.
  - -l: List all signals.
  - %N: Use a job number instead of a PID.



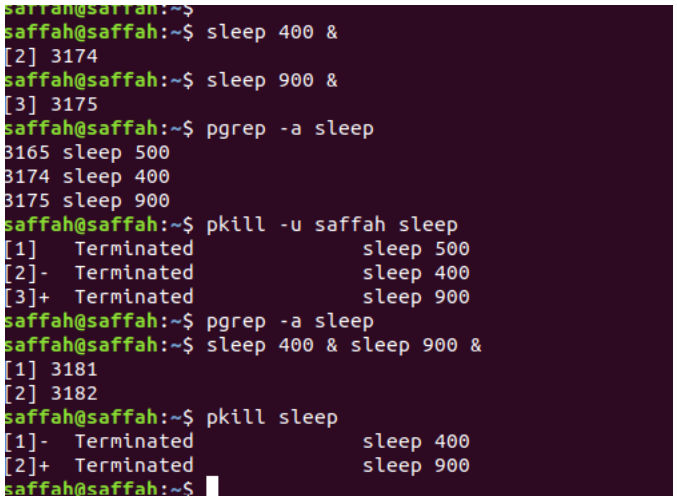
5.pkill

Pkill: It sends a signal (TERM by default) to processes matched by part of the name.

Syntax: pkill [-u USER] NAME

OPTIONS:

• -u USER: Only that user's processes.



6.killall

Killall: It sends a signal to every process with the exact name.

Syntax: killall NAME

OPTIONS:

• (none): Match the exact name and send SIGTERM.



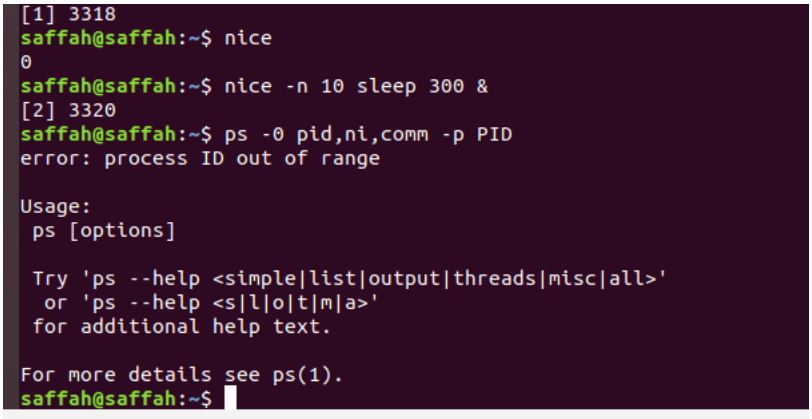
7.nice

Nice: It starts a command with a chosen niceness (priority). Niceness runs from -20 to 19. Higher is kinder to others. Default is 0.

Syntax: nice [-n N] command

OPTIONS:

- -n N: The niceness to start with.
- (none): Print your current niceness.
- -20 to -1: Greedier, only the administrator can set it.
- 1 to 19: Kinder, any user can set it.



8.renice

Renice: It changes the niceness of a process that is already running. You can always make your own process kinder; making it greedier needs sudo.

Syntax: renice -n N -p PID

OPTIONS:

- -n N: The new niceness.
- -p PID: Which process.



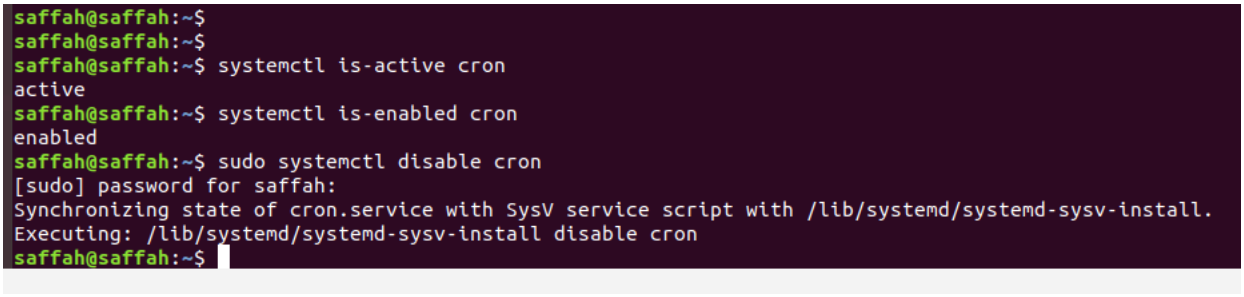
9. systemctl enable / disable

They decide whether the service starts at the next boot. They do not touch the running process unless you add --now.

Syntax: sudo systemctl enable|disable [--now] UNit.

OPTIONS:

- **enable**: Start at every boot.
- **disable**: Do not start at boot.
- --**now**: Also start or stop it immediately.



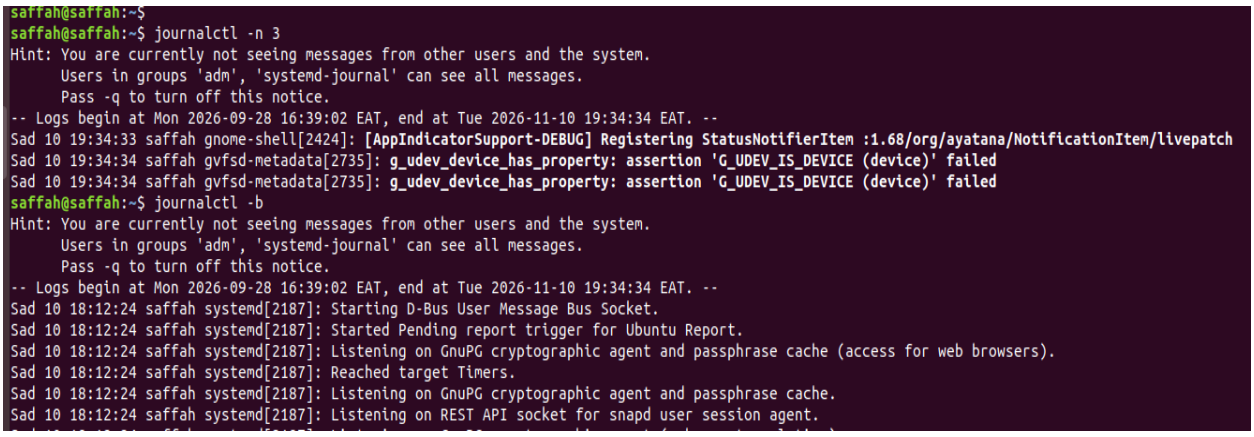
10. journalctl

Journalctl: It reads the systemd journal, which collects messages from the kernel and every service.

Syntax: journalctl [OPTION]

OPTIONS:

- -n N: The last N entries.
- -b: This boot only.
- -p err: Errors and worse.
- -u UNIT: One service only.
- --since "1 hour ago": From a start time.
- -xe: Jump to the end and add explanations.
- -f: Follow new lines live; Ctrl+C stops.
- -k: Kernel messages only.



11.logger

Logger: It writes a test line into the system log.

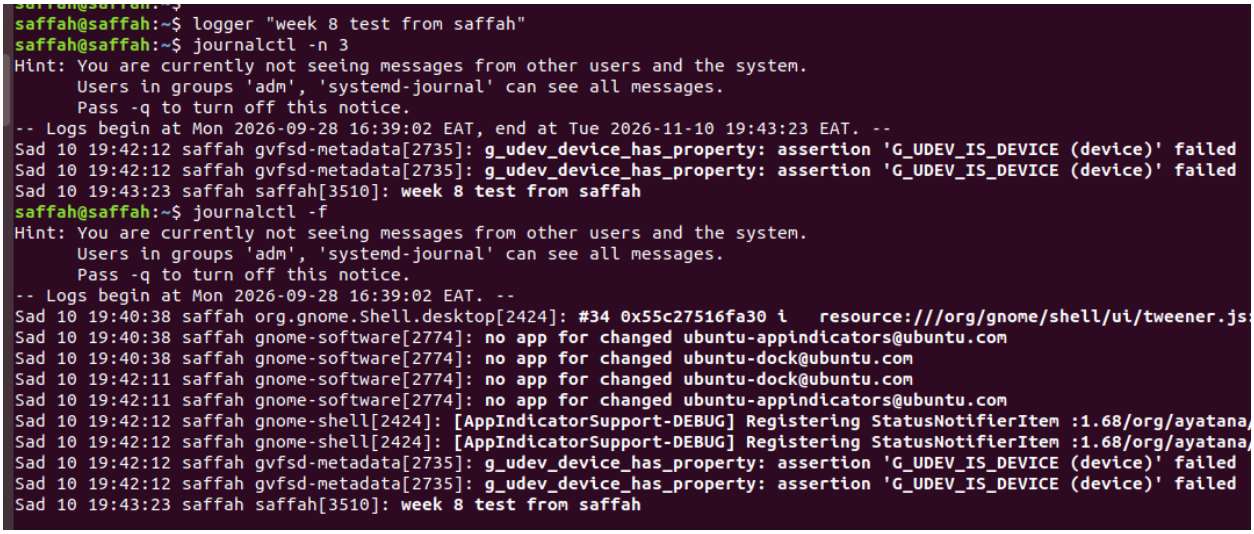
Syntax: logger "MESSAGE"

32./var/log (less, tail) /var/log: Classic text logs written by rsyslog. Read them with less and tail.

Syntax: less /var/log/FILE or tail [-n N] [-f] /var/log/FILE

OPTIONS:

- **syslog**: General system messages.
- **auth.log**: Logins, sudo, SSH.
- **kern.log**: Kernel messages.
- -n N (tail): Last N lines.
- -f (**tail**): Follow new lines.



12. shutdown, reboot, poweroff

- They shut the system down cleanly so services stop in order and the disk is written safely.

Syntax: sudo shutdown [-r] TIME

OPTIONS:

• now: Immediately.

• +N: In N minutes.

• -r: Reboot instead of power off.

• -c: Cancel a planned shutdown.

• reboot: Restart now.

• poweroff: Stop everything and switch off now.

